Master's degree programme Artificial Intelligence

Gehe zu:
Studieninhalt (/en/studies-at-thws/studying-at-thws/degree-programmes/artificial-intelligence/#studieninhalt)
Praxis (/en/studies-at-thws/studying-at-thws/degree-programmes/artificial-intelligence/#praxis)
Beruf und Perspektiven (/en/studies-at-thws/studying-at-thws/degree-programmes/artificial-intelligence/#beruf-perspektiven)

(https://www.thws.de/en/services-and-support/application-enrolment-exams-internship/application/)

Start of studies: 15 March

Apply now for a Master's degree programme!

For most of our master's degree programmes, you can apply by 15 January 2026. For English-taught Master's degree programmes not subject to admission restrictions (NC), the deadline is 15 December 2025.

Further information on applications

Facts and figures

Degree programme:
Artificial Intelligence

Degree:
Master of Science (M. Sc.)

Admission:
No admission restrictions

THWS location:
Würzburg, CAIRO and Campus Sanderheinrichsleitenweg

Faculty:
Computer Science and Business Information Systems (https://fiw.thws.de/en/)

Study mode:
Full-time degree programme

Duration:
3 semesters standard time to degree

Language:
English

Semester fees:
None (student services fee only (Studierendenwerksbeitrag); approx. EUR 150)

Start of studies:
Summer Semester (15 March)

Application period:
Annually, 15. October till 15. December

Contact

Contact person
Programme directors
Professor Dr.-Ing. Pascal Meißner and
Professor Dr. Andreas Lehrmann
mai.fiw[at]thws.de

Academic Advisory Service
studienberatung@thws.de

Degree programme website
Degree programme Artificial Intelligence (https://mai.thws.de/)

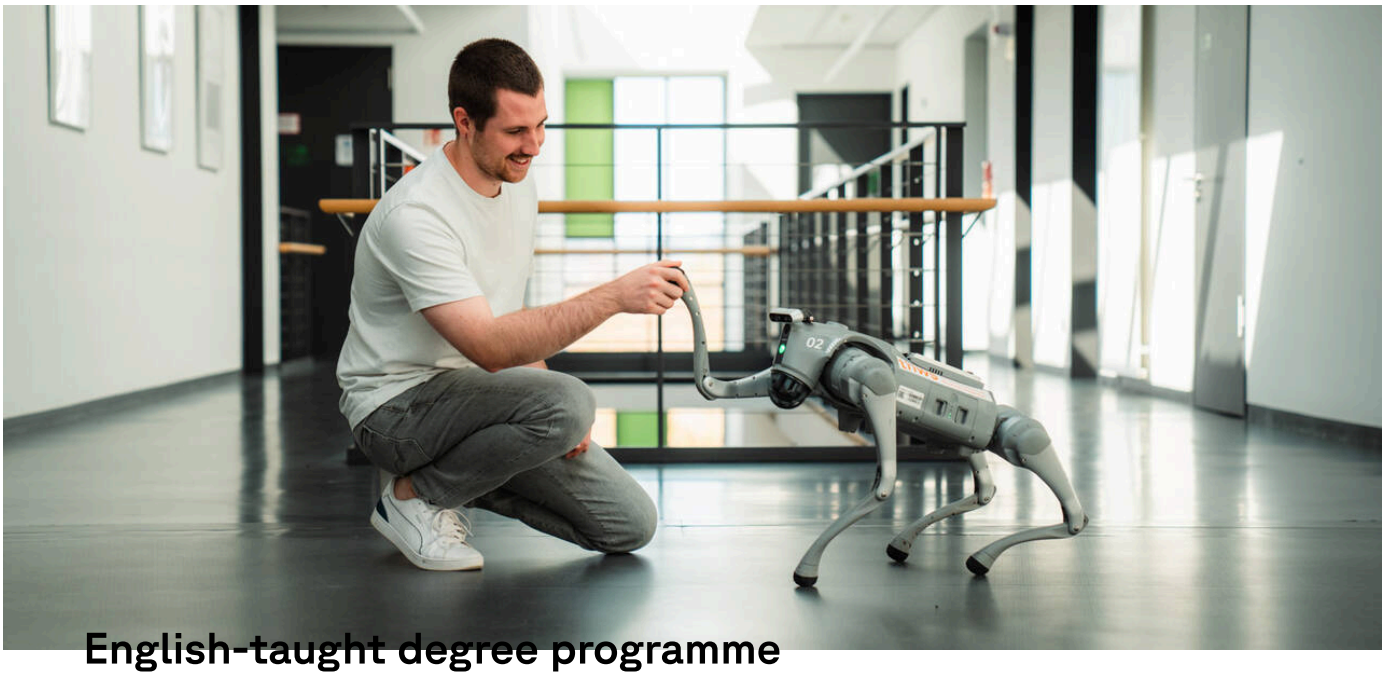
Information about applying (/en/services-and-support/application-enrolment-exams-internship/application/)

Apply now (https://campusportal.thws.de/qisserver/pages/cs/sys/portal/hisinoneStartPage.faces)

Information for internationals (http://international.thws.de/en/thws-international/ways-to-thws/incoming-full-time-students/application/)

Shape tomorrow through intelligent technology

Would you like to actively contribute to the digital transformation, understand data-driven systems, and develop new AI applications? The English-taught master's degree programme Artificial Intelligence at the Technical University of Applied Sciences Würzburg-Schweinfurt (THWS) offers you an in-depth, research-oriented education in the forward-looking field of artificial intelligence. Apply your analytical skills, broaden your methodological knowledge, and get ready for a career at the interface between science, industry, and society.



English-taught degree programme

Reasons to study Artificial Intelligence at THWS

Artificial Intelligence (AI) has made enormous progress over the last few years and has become an integral part of everyday life. The demand for qualified professionals in this field keeps growing. In industry in particular, sound skills and a systematic understanding of AI are required to effectively apply methods in practice. That is where the master's degree programme Artificial Intelligence comes in: you deepen your knowledge of the key principles and methods of AI, with a special emphasis on machine learning and mathematical foundations. You will also learn how to transfer and adapt these concepts to different fields of application of artificial intelligence.

Advantages of the degree programme:

- English-taught programme targeting international AI careers
- Your lecturers work in AI research and can thus transport the latest findings to their lectures
- Practice-oriented studies through team projects and early integration in ongoing research projects
- Individual focuses through electives, e.g. in the fields of autonomous systems, generative AI models, or entrepreneurship
- Ethics and law are a firm part of the curriculum
- Excellent supervision through small groups and personal exchange with the lecturers
- Diverse opportunities to go abroad due to our partner university network
- No study fees! Receive an excellent, English-taught education without the high fees usually charged in other countries

Three good reasons

At the pulse of research: all lecturers are actively involved in international AI research and contribute their recent findings to teaching

2.

Future-oriented studies: fundamental technologies and methods providing long-term value are integral parts of the modules

3.

Study without fees: benefit from an excellent, English-taught degree programme without high costs

Study contents

The three-semester master's degree programme combines theoretical knowledge with practical application. The focus is on methods of artificial intelligence, in particular machine learning and mathematical foundations. You will learn to critically question existing solutions, apply existing knowledge to new and complex issues, and solve problems systematically and methodically. Elective modules allow for individual specialisation. Practical projects and early involvement in AI laboratories promote independent academic work and prepare you optimally for your future career or doctorate.

Programme structure

1: Main studies (1st and 2nd semester)

In the first semester, you will lay the methodological and mathematical foundations for your in-depth understanding of artificial intelligence, with a particular focus on machine learning. A scientific seminar will help you to systematically address your own research questions. In the second semester, the focus will be on specialised areas of application such as robotics, language processing, and image processing, supplemented by ethical and legal dimensions. In an extensive practical project, you will work in a team on complex problems, apply your knowledge and further develop your ability to critically analyse and structurally solve real-world challenges.

2: Final phase (3rd semester)

In the third semester, you will independently work on a scientific question as part of your master's thesis. In doing so, you will demonstrate that you can evaluate existing solutions in a well-founded manner, transfer existing knowledge to new contexts and work on challenging problems in a structured manner

with in-depth methodological knowledge. You may also write your master's thesis in cooperation with a company or at an international partner university. You can hone your profile with additional elective modules.

Specialisations

Specialisations provide you with the opportunity to choose an individual focus for your studies. Among others, we offer the following elective modules:

- Computational Creativity
- Deep Generative Modelling
- Entrepreneurship for Engineers

How are studies at THWS particularly practice-oriented?

- Continuous practical exercises in teaching
- Practical projects and early involvement in research practice
- Professors with extensive experience in industry and economy
- Cooperation with companies and research institutes, also internationally
- Promotion of final theses written in cooperation with a company

Career and perspectives

There is a huge demand for AI experts in all areas of application in industry and research. With your master's degree in Artificial Intelligence, you will have numerous career opportunities in AI, data science, cognitive computing, and many other exciting fields. Since artificial intelligence is an interdisciplinary field, the programme focuses on a broad basis. You will gain knowledge in the fields of computer science, mathematics, physics, and cognitive science and expand your range of skills so that you are well equipped to meet the diverse requirements of AI. After graduation, you will be ideally prepared to advance the development of AI systems in science, research, or industry. You will also qualify for management positions in which you can take responsibility for the further development, integration, and secure operation of AI technologies.

Possible professional fields include:

- Development of AI systems and intelligent applications
- Machine learning and deep learning
- Data science and big data analytics
- Cognitive computing
- Research and development in companies or academic institutions
- Software development in AI and data-based systems
- IT consulting and technology management
- Automation and intelligent production processes (e.g. Industry 4.0)
- Product and project management in technology
- Ethical analysis, regulation and governance of AI systems

Is Artificial Intelligence the right master's degree programme for you?

If you have good to very good knowledge and skills in computer science, proven through a relevant theory- and practice-based academic qualification, we would like to encourage you to apply for the master's programme in Artificial Intelligence.

The English-taught, research-oriented master's programme offers you the opportunity to deepen the knowledge from your studies in computer science or a related discipline. You will develop application-oriented and methodological skills in the field of AI and learn how to conduct academic work independently in an academic manner. Upon graduation, you will be able to choose and apply suitable AI approaches. Your knowledge will enable you to develop intelligent systems and tackle new, practice-oriented issues. The programme is particularly suitable if you would like to work on future-oriented technologies in an international, English-speaking environment.

Admission requirements

To be admitted to the programme, a bachelor's degree with 210 ECTS and an overall grade of 2.5 or better in the field of computer science from a German university or an equivalent degree is required.

In order to apply for this programme, you will additionally need to submit the application form available from the website of the Faculty of Computer Science and Business Information Systems (https://mai.thws.de/en/).

For more detailed information on the admission requirements of this master's degree programme, please see the website of the faculty in charge. Specific requirements are also indicated in the programme's Study and Examination Regulations.

Please also see the language requirements (/en/services-and-support/application-enrolment-exams-internship/admission-and-enrolment/language-entrance-requirements/).

Your interests and abilities

- Enthusiasm for artificial intelligence and its applications
- Good or very good knowledge of mathematics and computer science
- Interest in independent, methodological work
- Ready-to-use, advanced programming skills, ideally using Python
- Willingness to conduct interdisciplinary project work
- Academic thinking and structured problem-solving
- Interest in ethical and legal issues surrounding digital technologies
- Motivation to learn and work in English in an international environment

Application

Apply from 15 October to 15 December via our application portal to start your studies in the upcoming summer semester.

To the step-by-step guide (/en/services-and-support/application-enrolment-exams-internship/application/process-in-thws-campusportal/)

To the application portal (https://campusportal.thws.de/qisserver/pages/cs/sys/portal/hisinoneStartPage.faces)

You have already completed study achievements and thus want to apply for a higher semester? Contact our Department of Student Affairs.

Information on applying for a higher semester (/en/services-and-support/application-enrolment-exams-internship/application/applying-for-higher-semesters/)

Do you have unanswered questions?

You can contact the programme advisor, i.e. a professor of the field of study you are interested in and can ask them your questions.

To the programme advisors (/en/services-and-support/academic-advisory-service/programme-related-information-and-advice/)

If you are unsure which degree programme would suit you best, or in case of general questions, please do not hesitate to contact the Academic Advisory Service.

To the Academic Advisory Service (/en/services-and-support/academic-advisory-service/)

During the THWS Taster Sessions, you can attend a variety of lectures in the degree programme of your choice and get valuable first impressions of life on campus.

To the THWS Taster Sessions (/en/prospective-students/taster-sessions/)

During the Student Information Days in March, you can directly exchange with students and professors and gain information on the degree programmes you are interested in.

To the student information days (/en/prospective-students/student-information-days/)

Get informed

Looking for more information? For the most important information on studying at THWS, check out the overview.

Study overview (/en/prospective-students/)

Quicklinks

Dates and events (/en/studiengangs-db-neu-en/footer/dates-and-events/)

FAQ for prospective students (/en/prospective-students/faq-for-prospective-students/)

Getting started (/en/studiengangs-db-neu-en/footer/getting-started/)

Application, enrolment, exams, internship (/en/studiengangs-db-neu-en/footer/application-enrolment-exams-internship/)

Information for pupils (/en/studiengangs-db-neu-en/footer/information-for-pupils/)

Change of studies (/en/studiengangs-db-neu-en/footer/change-of-studies/)

THWS International (https://international.thws.de/en/)

Contact Service Center

Würzburg

Münzstraße 12
97070 Würzburg

Phone +49 931 3511-0 (tel:+49 931 3511-0)
E-Mail servicezentrale-wue[at]thws.de

Map (https://www.google.com/maps/place/M%C3%BCnzstra%C3%9Fe%2012%0A97070%20W%C3%BCrzburg)

Schweinfurt

Ignaz-Schön-Straße 11
97421 Schweinfurt

Phone +49 9721 940-5 (tel:+49 9721 940-5)
E-Mail servicezentrale-sw[at]thws.de

Map (https://www.google.com/maps/place/Ignaz-Sch%C3%B6n-Stra%C3%9Fe%2011%0A97421%20Schweinfurt)

Faculties

- Applied Natural Sciences and Humanities (http://fang.fhws.de)
- Applied Social Sciences (http://fas.fhws.de/en/home_page.html)
- Architecture and Civil Engineering (http://fab.fhws.de/#)
- Business and Engineering (https://fwi.fhws.de/en/)
- Computer Science and Business Information Systems (http://fiw.fhws.de/#)
- Economics and Business Administration (http://fwiwi.fhws.de/#)
- Electrical Engineering (http://fe.fhws.de/#)
- Mechanical Engineering (http://fm.fhws.de/#)
- Plastics Engineering and Surveying (https://www.fhws.de/fakultaeten/kunststofftechnik-und-vermessung/)
- Visual Design (http://fg.fhws.de/#)